

Overview

Causes

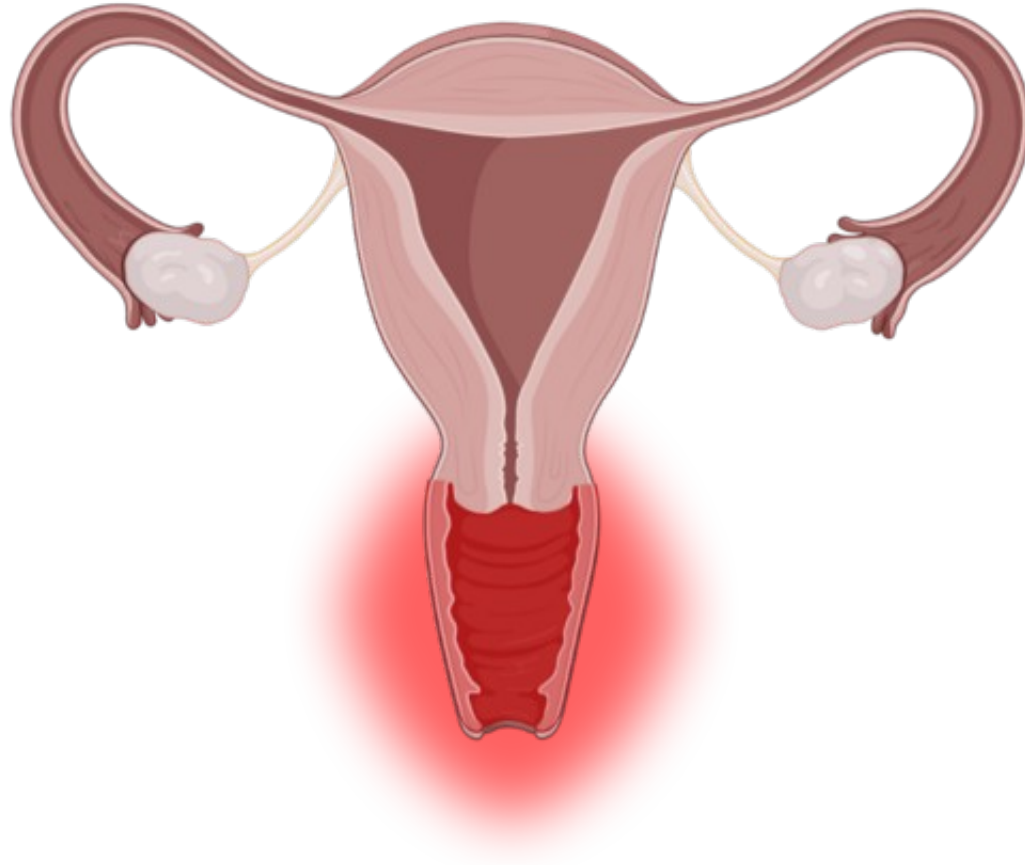
Evaluation

Management

Cases

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Vaginitis



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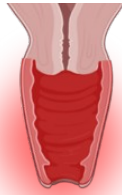
Cases

Symptoms

Causes

Mimics

Overview



Itching
Burning



Abnormal discharge
Abnormal odor

Infectious



Bacterial Vaginosis



Candidal vulvovaginitis



Trichomonal vaginitis

Noninfectious

Atrophic vaginitis/ GU syndrome
of menopause (GSM)

Irritant vaginitis

Allergic vaginitis

Inflammatory vaginitis

STIs (gonorrhea, chlamydia) can cause cervicitis which mimic
symptoms of vaginitis



Overview

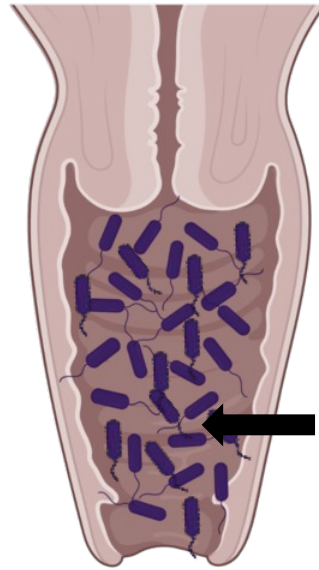
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Pathophysiology of Vaginitis



70-90% *Lactobacillus*

Produces lactic acid →
pH 3.8-4.5

Estrogen promotes
lactobacilli concentrations

Normal
Vaginal Flora

Lactobacillus helps regulate
vaginal pH and inhibits
other bacterial growth

Hormonal fluctuations
dictate vaginal flora and pH

BV

Yeast

Trich

GSM

Overview

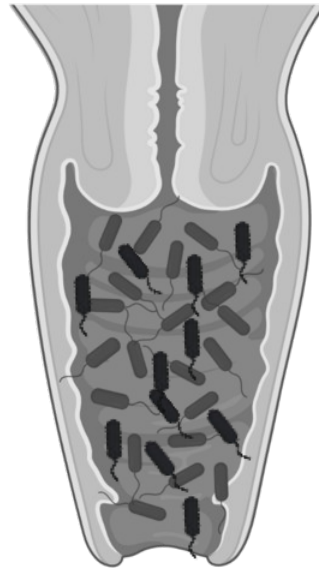
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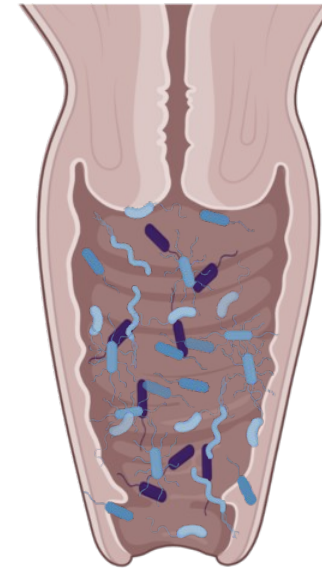
Pathophysiology of Vaginitis



70-90% *Lactobacillus*

Produces lactic acid →
pH 3.8-4.5

Estrogen promotes
lactobacilli concentrations



↓ *Lactobacillus*

↓ lactic acid → ↑ pH > 4.5

Normal
Vaginal Flora

BV

Shift in vaginal flora
(*Gardnerella* or
Mycoplasma hominis)
leads to ↓ *Lactobacillus*

Most common cause of
vaginitis (15-50%)

Yeast

Trich

GSM

Overview

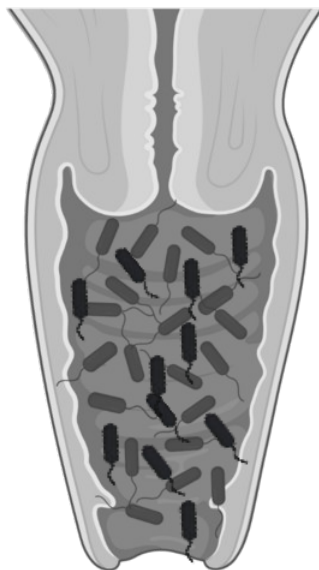
Causes

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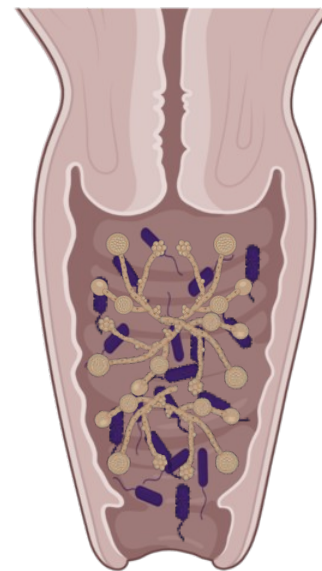
Pathophysiology of Vaginitis



70-90% *Lactobacillus*

Produces lactic acid →
pH 3.8-4.5

Estrogen promotes
lactobacilli concentrations



Overgrowth of *Candida*
(no change in lactobacilli)

No shift in vaginal pH

Normal
Vaginal Flora

BV

Yeast

Trich

GSM

Overgrowth of *Candida albicans* (80-92% of cases), *krusei*, or *glabrata* + ↓ in host immune regulation

75% of females will have at least 1 episode

Overview

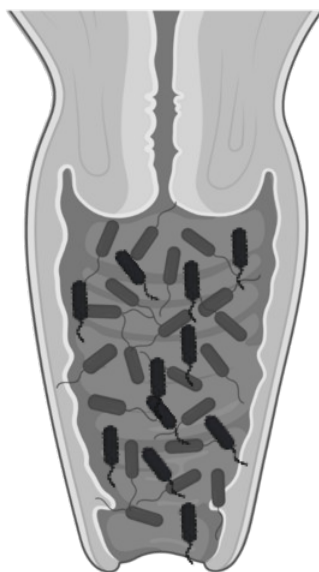
Causes

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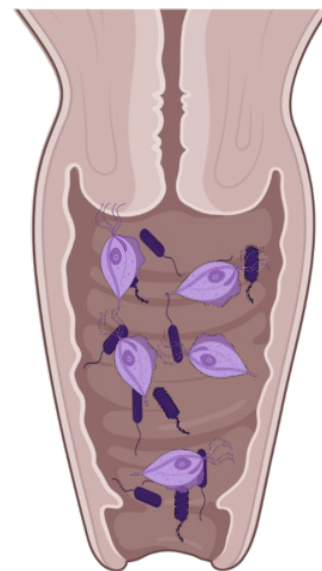
Pathophysiology of Vaginitis



70-90% *Lactobacillus*

Produces lactic acid →
pH 3.8-4.5

Estrogen promotes
lactobacilli concentrations



↓ *Lactobacillus*

↓ lactic acid → ↑ pH > 4.5

Normal
Vaginal Flora

BV

Yeast

Trich

GSM

Infection with motile
protozoan *Trichomonas*
vaginalis

Highly contagious STI
5-50% of acute vaginitis
Incubation of 4-28 d

Overview

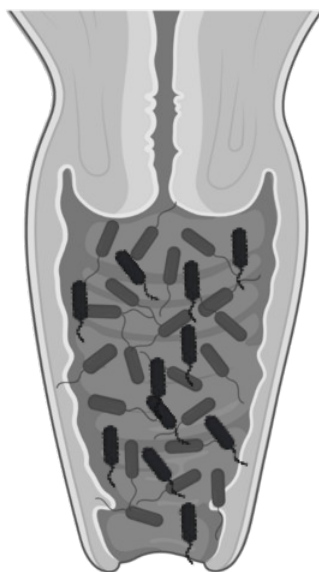
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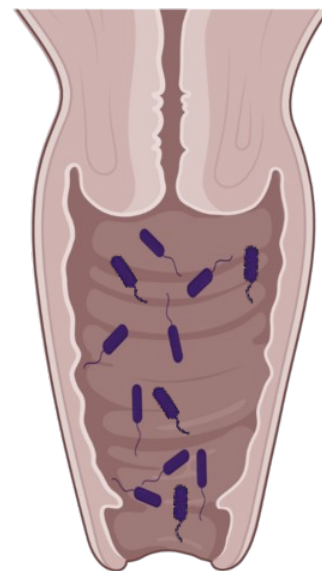
Pathophysiology of Vaginitis



70-90% *Lactobacillus*

Produces lactic acid →
pH 3.8-4.5

Estrogen promotes
lactobacilli concentrations



↓ *Lactobacillus* (from ↓
estrogen)

↓ lactic acid → ↑ pH > 5.0-5.5

Normal
Vaginal Flora

BV

Yeast

Trich

GSM

↓ estrogen in
menopause results in
↓ *Lactobacillus*

Evaluation – History and Physical

Overview

Causes

Evaluation

H&P

Labs

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BV

Yeast

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Risk factors

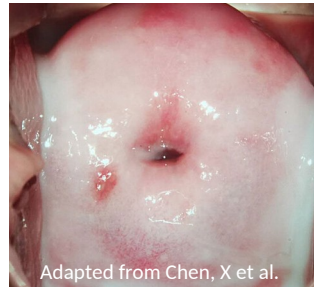
- New or multiple partners
- Same-sex partners
- Douching, smoking

- Antibiotic use
- Diabetes
- Luteal phase (menstrual cycle)
- Immunosuppression

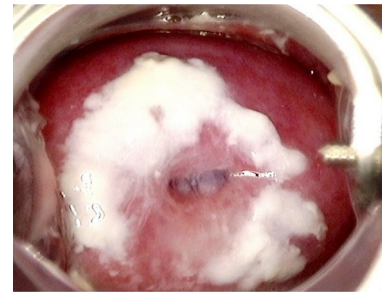
- New or multiple partners
- Smoking
- No barrier protection, other STIs

Vaginal discharge

Thin, homogenous
Clear, white, or gray



Thick, “curd-like” white, non-odorous



Copious, frothy, green-yellow

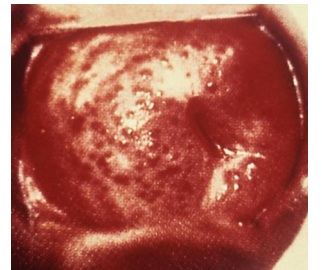


Other signs and symptoms

- No inflammation
- ± fishy odor

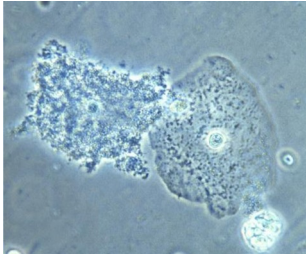
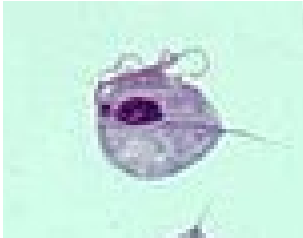
- Erythematous labia and vulva, excoriations
- Burning, itching

- Vulvar erythema, edema
- “Strawberry” cervix



Evaluation - Lab Testing

- Overview
- Causes
- Evaluation
 - H&P
 - Labs
- Management
- Cases
 - Discharge + KOH → Fishy odor

	BV	Yeast	Trich
pH	pH > 4.5 Sn 97% Sp 64%	Normal, pH 4.0 to 4.5	pH > 4.5 (often >5) Sn 56% Sp 50%
Whiff test	Positive Sn 43% Sp 91%	Negative	May be positive
Wet prep	Clue cells with saline Sn 80% Sp 80% 	Hyphae/ budding (KOH) Sn 50-70% Sp 97%  From Wikimedia commons.	Motile trichomonads, ↑ WBC Sn 50-60% Sp 95% 
Diagnostic tests	- Amsel criteria - Gram stain - DNA assay VPIII (Sn 90%, Sp 96-97%)	- Culture (3-4+ growth) - DNA assay VP III - PCR (Sn 90%, Sp 94%) - Pap smear	- DNA assay VP III - NAAT PCR (Sn 99%, Sp 99%) - Rapid antigen testing - Pap smear
Most Sensitive?			

Unless otherwise specified, images adapted from CDC Public Health Image Library.

- Overview
- Causes
- Evaluation
- Management
- Cases

Management

	BV	Yeast	Trich
Topical Tx	 Topical Tx?	 Mid Miconazole vaginal supp x 1 Clotrimazole x 3d Topical Tx?	Topical Tx?
Oral Tx	 Metronidazole 500 mg BID x Clindamycin 300 mg BID x 7d Oral Tx?	Flu Oral Tx?	 Metronidazole 500mg BID x Tinidazole 2g x 1 Oral Tx?
Prevention	- Fewer partners - A - p Prevention? Sexual devices Smoking cessation	- Glycemic control - Avoid Prevention? iotics	- Fewer partners - Bar - Pro Prevention? ermicide al devices Avoiding douching

Overview

Causes

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Cases

1

2-3

Case 1a

23-year-old F with no PMH presents to PCP's office with watery, malodorous discharge.

What additional history is needed to help with diagnosis?

- Discharge: consistency, color, odor
- Associated sx: pruritus, burning, irritation, dyspareunia, abdominal pain, nausea
- Sexual history
- Menstrual history

...Meds: combined OCPs

ROS: copious white-gray discharge, no itching, pelvic pain, and dyspareunia.

Menstrual hx: LMP ~5 wks ago.

Sexual hx: sexually active with 1 male partner. No douching, barrier contraception, sex toys or spermicides

What is your differential for her symptoms?

- Vaginitis (infectious): BV, candidal, trichomonas vaginitis
- Cervicitis - gonorrhea ± chlamydia
- Vaginitis (non-infectious): atrophic vaginitis/ GSM, allergic/ irritant vaginitis
- Erosive lichen planus

Case 1b

...She otherwise eats a well-balanced diet and denies any substance use.

What are her risk factors for vaginitis?

- No barrier contraception ↑ risk of STIs
- Combined OCPs ↑ physiologic discharge
- No risk factors for BV (douching, smoking, same-sex or multiple partners) or candidal infection (high sugar diet, DM, immunosuppression, recent antibiotics)

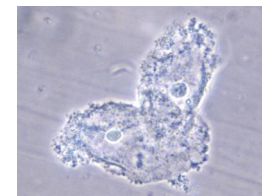
What is your next step in evaluation?

- Perform a **pelvic exam** and send additional testing - **pregnancy test, wet mount, pH, KOH prep, ± STI**. Consider ± gram stain, culture, DNA assay and NAAT if available.
- Symptoms of vaginitis are nonspecific. History cannot reliably differentiate between causes.

Case 1c

Pelvic exam: moderate, thin white discharge with a faint fishy odor. Cervix without erythema, tenderness, or discharge.

Wet mount:



Vaginal pH: 5.0

KOH + discharge produces fishy odor.

Pregnancy test: positive

What does the wet mount show?

Clue cells - vaginal epithelial cells with ruffled borders from adherent bacteria

What is the most likely diagnosis?

BV. She meets **Amsel criteria** (3/4 for dx):

1. Thin homogenous white discharge
2. >20% clue cells on wet mount
3. Positive Whiff test
4. Vaginal pH > 4.5

How will you treat her symptoms?

Flagyl 500 mg PO BID x 7 d OR Flagyl 0.75% 5 g intravaginal qD x 5 d.

Overview

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Case 2

36 year-old F presents with thick, white vaginal discharge x days. Also reports itching and "irritated" vulva. Sexually active with 1 female partner.

Pelvic exam: Vulvar erythema and thick, white discharge. No cervical discharge. No cervical or adnexal tenderness.

Vaginal pH: 4

Wet prep:



What does the wet prep show?

Hyphae and budding yeast.

What is the most likely diagnosis?

Candidal vulvovaginitis.

Are same sex partners at risk for vaginitis?

↑ risk of BV. *Trichomonas* possible with shared sex toys.

What is next best step in evaluation or management?

- Empiric tx: PO fluconazole 150 mg x 1 or vaginal suppositories.
- Additional testing for vaginitis (culture, DNA assay, Whiff test).
- Screen for STIs (HIV), DM, and recent antibiotic use.

Case 3

24 year-old F presents with vaginal discharge x 2 wk. LMP ~ 3 weeks ago. Sexually active with 1 new male partner x 1 month and intermittently uses condoms. Recently started douching.

Pelvic exam: cervix appears erythematous. No pelvic or cervical tenderness on exam. The discharge is thick and yellow.

Wet prep: clue cells

Pregnancy test: negative

Vaginal pH is > 5

Whiff test: positive

What is your next step in evaluation or management?

- Additional testing for STIs, DNA assay or PCR for *Trichomonas* (wet prep is only 50-60% sensitive)
- Consider empiric treatment of BV and *Trichomonas* with metronidazole 500 mg PO BID x 7 d. Note topical treatments will not treat *Trichomonas*.

...DNA assay (VP11) returns + for BV and *Trichomonas*. Gonorrhea and chlamydia are negative. HIV negative.

What additional counseling would you provide?

- Treat sex partner (metronidazole 2 g PO x 1 for men)
- Avoid sexual contact until symptoms resolve
- Test for re-infection after 3 months
- Encourage condom use/ barrier contraception
- Avoid douching and vaginal cleansers to reduce risk of recurrence

Newer Context (2021–2025) — core teaching unchanged

- **Diagnosis & first-line treatment are unchanged: deck regimens match the 2021 CDC STI Treatment Guidelines (still current) — BV/Trich: metronidazole 500 mg PO BID ×7 d.**
- Trichomoniasis in women: multi-dose metronidazole (500 mg BID ×7 d) replaced single 2 g dose as preferred (more effective; Kissinger, 2021 CDC) — already reflected on the deck. Single 2 g dose still used for partners/men.
- Secnidazole (Solosec) 2 g oral granules ×1 — single-dose option, FDA-approved for BV (2017) and *T. vaginalis* (2021); only single-dose oral agent for both.
- Recurrent vulvovaginal candidiasis (RVVC) — two new oral agents: ibrexafungerp (Brexafemme, first-in-class triterpenoid — VVC 2021, RVVC 2022) and oteseconazole (Vivjoa, FDA 2022, first drug approved specifically for RVVC).
- GSM/atrophic vaginitis: vaginal estrogen, moisturizers/lubricants; non-hormonal options for those avoiding estrogen.
 - Sources: CDC MMWR 2021 (RR-70/4); ACOG PB 215; FDA approvals — Solosec 2017/2021, Brexafemme 2021/22, Vivjoa 2022.

Take-Home Points

- The three common infectious causes disrupt vaginal flora differently → predictable pH: BV (anaerobe overgrowth, ↓ lactobacilli, pH >4.5), yeast (Candida overgrowth, NORMAL pH 4.0–4.5), trichomonas (↓ lactobacilli, pH >4.5, often >5).
- Symptoms alone cannot distinguish causes — offer an exam with point-of-care pH + microscopy (KOH/wet prep) and/or DNA assay/NAAT to all symptomatic patients; exclude STI cervicitis (GC/chlamydia) mimics.
- Key bedside clues: clue cells + positive whiff + Amsel ≥3/4 → BV; hyphae/budding on KOH + normal pH → yeast; motile trichomonads (only 50–60% sensitive — confirm with NAAT) → trichomonas.
- Treat by etiology: BV → metronidazole (PO/topical) or clindamycin; yeast → fluconazole 150 mg PO ×1 or topical azole; trichomonas → metronidazole 500 mg BID ×7 d AND treat partner(s). Topical azoles preferred in pregnancy.
- Address risk reduction for recurrence (avoid douching, clean sex devices, barrier protection, glycemic control, smoking cessation); consider newer RVVC agents for recurrent yeast.

Attribution & Changelog

- Original: "Vaginitis," TeachIM.org, published Dec 2022 — Rachel S. Casas MD EdM; Section Ed. Adelaide McClintock MD; Exec. Ed. Yilin Zhang MD.
 - https://teachim.org/teaching_material/vaginitis/ — CC BY-NC 4.0 (attribution required, non-commercial). Optimized June 2026. Deck: Vaginitis-1-yz-2.6.pptx (from page embed URL).
- **Corrections to this copy (all click-to-reveal animations & BioRender graphics preserved):**
 - Slide 5: Candida "kruseii" → "krusei" (correct species name; deck flagged it as a spelling error).
 - Slide 11 (Case 1): "What your differential for her symptoms?" → "What is your differential..." (grammar).
- Appended: this changelog + Newer-Context (2021–25) + Take-Home slides. No original content removed; 12 original slides → 15.
- Companions copied & linked (reviewed, no changes): Vaginitis-Teaching-Instructions-12_13_22.pdf (teaching guide), Learner-Handout-vaginitis-PDF.pdf, Learner-Handout-vaginitis.pptx. Package: HTML, OnePager PDF, Anki TSV.
- Educational use only — not medical advice; clinician-review before teaching.